

Sheet 6: Learning to Rank

Exercise 1 (Bayesian Personalized Ranking)

[10 Points]

In the previous tutorial, we trained matrix factorization models for rating prediction. We now shift the focus from predicting ratings to optimizing item rankings.

In this exercise, we will use the MovieLens dataset together with the **Cornac framework** to train and evaluate ranking-based recommender systems using Bayesian Personalized Ranking (BPR).

- a) **0 Points** Convert the MovieLens dataset into implicit feedback by defining which interactions should count as positive feedback (e.g., ratings ≥ 3).
- b) **1 Point** Discuss the implications of treating missing interactions as unknown or negative feedback, and explain why implicit recommendation settings are fundamentally different from explicit rating prediction.
- c) **3 Points** Train a BPR model using Cornac. Explain how the model represents users and items through latent embeddings and how pairwise ranking is optimized during training.
- d) **3 Points** Experiment with several hyperparameters such as the embedding dimensionality, learning rate, regularization strength, and number of training epochs. Analyze how these choices influence recommendation quality and training behavior.
- e) **2 Points** Generate top- k recommendations for each user and evaluate them using ranking-oriented metrics such as Precision@ k , Recall@ k , MAP@ k , or nDCG@ k . Additionally, compare the BPR model to a standard matrix factorization model from the previous tutorial. Discuss how the optimization objective changes the learned recommendations.
- f) **1 Point** Reflect on the relationship between learning objectives and evaluation metrics. In which scenarios is rating prediction preferable, and in which scenarios are ranking-based approaches more appropriate?

Important: Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.